MILITARY SPECIFICATION

SEMICONDUCTOR DEVICES, THYRISTORS
TYPES 2N1870A, 2N1871A, 2N1872A AND 2N1874A

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon thyristors and is in accordance with MIL-S-19500, except as otherwise specified herein. The units have the following characteristics.

1.2 Maximum ratings[1/].

I_o T_A = 100° C	i_f (surge)	V_G V_{CG}	I_G	i_G	Operating and storage temperature	Operating altitude
mAdc	a	Vdc	mAdc	ma	°C	mm Hg
220	15	5	25	250	-65 to +150	15

[1/] For additional information on ratings see figures 1 and 2.

Type	V_{RM}[1/]	V_{FBXM}[1/]
	v(pk)	v(pk)
2N1870A	30	30
2N1871A	60	60
2N1872A	100	100
2N1874A	200	200

[1/] Blocking voltage ratings apply with gate connected through a resistor, R_2 = 1k ohm or less, to cathode as shown on figure 3.

1.3 Primary electrical characteristics.

Limits	V_f i_f = 2A	I_H	V_{GT} V_{AC} = 5V R_e = 100 ohms	I_{GT} V_{AC} = 5V R_e > 10K
	V	mAdc	Vdc	µAdc
Min.	---	0.3	0.4	---
Max	2.5	5.0	0.8	200

FSC 5961

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500 and as follows:

I_{FBX}	–	DC forward blocking current with resistor R_2 between gate and cathode
I_{RBX}	–	DC reverse blocking current with resistor R_2 between gate and cathode
i_G	–	Peak gate current
V_{AA}	–	Anode supply voltage
V_{FBX}	–	Forward anode voltage
V_{CG}	–	Reverse gate voltage
V_G	–	Rated reverse gate voltage

3.3 Design, construction, and physical dimensions. Thyristors shall be of the design, construction, and physical dimensions shown on figure 4.

3.4 Performance characteristics. The performance characteristics of the thyristors shall be as specified in tables I and II.

3.5 Marking. Thyristors shall be marked in accordance with MIL-S-19500, except that the following may be omitted at the option of the manufacturer:

(a) Country of origin
(b) Manufacturer's identification

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I and II.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A and B inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

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4.3.3 Quality conformance inspection information. Upon request, one copy of the quality conformance inspection data pertinent to the inspection lot shall accompany the shipments.

TABLE I. Group A inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			5				
Visual and mechanical examination	2071			---	---	---	---
Subgroup 2			5				
Forward blocking current	4206	DC method, bias cond. B; R_2 = 1,000 ohms, $\frac{1}{4}$ watt		I_{FBX}	---	10	µA
2N1870A		V_{FBX} = 30					
2N1871A		V_{FBX} = 60					
2N1872A		V_{FBX} = 100					
2N1874A		V_{FBX} = 200					
Reverse blocking current	4211	DC method, bias cond. B; R_2 = 1,000 ohms, $\frac{1}{4}$ watt		I_{RBX}	---	10	µA
2N1870A		V_R = 30					
2N1871A		V_R = 60					
2N1872A		V_R = 100					
2N1874A		V_R = 200					
Gate trigger voltage	4221	$V_2 = V_{FBX}$ = 5V R_e = 1,000 ohms R_L = 100 ohms		V_{GT}	0.4	0.8	V
Gate trigger current	4221	$V_2 = V_{FBX}$ = 5V R_e = 1,000 ohms R_L = 100 ohms		I_{GT}	---	200	µA
Forward on-voltage	4226	i_{FM} = 4 a(pk) pulse; pulse width = 8.5 msec max; duty cycle = 2% max		v_f	---	2.5	V
Exponential rate of voltage rise	4231	Bias cond. B; repetition rate = 60 pps; 50Ω ≤ R_L ≤ 400Ω; C = .001 µfd min; test duration = 15 sec; dv/dt = 100 V/µsec; R_3 = 1,000 ohms		V_{FBX}	---	---	---
Reverse gate current	4219	V_{CG} = 5V, anode open		I_{KG}	---	10	µA

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TABLE I. Group A inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Holding current	4201	I_G = -150 µA V_{AA} = 5V		I_H	0.3	5	mA
Subgroup 3			5				
High temperature operation		T_A = 125° C					
Forward blocking current	4206	R_2 = 1,000 ohms		I_{FBX}	---	100	µA
2N1870A		V_{FBX} = 30					
2N1871A		V_{FBX} = 60					
2N1872A		V_{FBX} = 100					
2N1874A		V_{FBX} = 200					
Reverse blocking current	4211	R_2 = 1,000 ohms		I_{RBX}	---	100	µA
2N1870A		V_R = 30					
2N1871A		V_R = 60					
2N1872A		V_R = 100					
2N1874A		V_R = 200					
Gate trigger voltage	4221	V_2 = V_{FBX} = 5V R_e = 1,000 ohms R_L = 100 ohms		V_{GT}	0.2	---	V
Holding current	4201	I_G = -150 µA V_{AA} = 5V		I_H	0.2	---	mA
Subgroup 4			5				
Low temperature operation		T_A = -65° C					
Gate trigger voltage	4221	V_2 = V_{FBX} = 5V R_e = 1,000 ohms R_L = 100 ohms		V_{GT}	---	1.0	V
Gate trigger current	4221	V_2 = V_{FBX} = 5V R_e = 1,000 ohms R_L = 100 ohms		I_{GT}	---	500	µA
Holding current	4201	I_G = -150 µA V_{AA} = 5V		I_H	---	15	mA

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TABLE II. Group B inspection

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 1			10				
Physical dimensions	2066	---		---	---	---	---
Subgroup 2			10				
Solderability	2026	---		---	---	---	---
Thermal shock (temperature cycling)	1051	Test cond. F					
Moisture resistance	1021	---		---	---	---	---
End points:							
Forward blocking current	4206	DC method, bias cond. B; R_2 = 1,000 ohms, $\frac{1}{4}$ watt		I_{FBX}	---	10	µA
2N1870A		V_{FBX} = 30					
2N1871A		V_{FBX} = 60					
2N1872A		V_{FBX} = 100					
2N1874A		V_{FBX} = 200					
Reverse blocking current	4211	DC method, bias cond. B; R_2 = 1,000 ohms, $\frac{1}{4}$ watt		I_{RBX}	---	10	µA
2N1870A		V_R = 30					
2N1871A		V_R = 60					
2N1872A		V_R = 100					
2N1874A		V_R = 200					
Gate trigger current	4221	V_R = 5V R_e = 1,000 ohms R_L > 10,000 ohms		I_{GT}	---	200	µA
Gate trigger voltage	4221	$V_2 = V_{FBX}$ = 5V R_e = 1,000 ohms R_L = 100 ohms		V_{GT}	0.4	0.8	V
Subgroup 3			10				
Shock	2016	Nonoperating; 1,000 G; 1.0 msec; 5 blows in orientation: Y_1, Y_2, and Z_1.					
Vibration fatigue	2046	---		---	---	---	---
Constant acceleration	2006	25,000 G		---	---	---	---
Vibration, variable frequency	2056	---		---	---	---	---

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TABLE II. Group B inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
End points: (Same as for subgroup 2)							
Subgroup 4			20				
Lead fatigue	2036	---		---	---	---	---
Subgroup 5			10				
Barometric pressure reduced, 2N1874A only	1001	15 mm Hg V_{FBX} = -200 Vdc 60 seconds					
Subgroup 6			20				
Salt atmosphere	1041			---	---	---	---
End points: (Same as for subgroup 2)							
Subgroup 7			10				
Surge current	4066	i(surge) = 15a peak I_o = 250 mA		---	---	---	---
End points:							
Forward blocking current	4206	R_2 = 1,000 ohms		I_{FBX}	---	10	µA
2N1870A		V_{FBX} = 30					
2N1871A		V_{FBX} = 60					
2N1872A		V_{FBX} = 100					
2N1874A		V_{FBX} = 200					
Reverse blocking current	4211	R_2 = 1,000 ohms		I_{RBX}	---	10	µA
2N1870A		V_R = 30					
2N1871A		V_R = 60					
2N1872A		V_R = 100					
2N1874A		V_R = 200					
Gate trigger current	4221	$V_2 = V_{FBX}$ = 5V R_e = 1,000 ohms R_L > 100 ohms		I_{GT}	---	200	µA
Gate trigger voltage	4221	$V_2 = V_{FBX}$ = 5V R_e = 1,000 ohms R_L = 100 ohms		V_{GT}	0.35	0.85	V

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TABLE II. Group B inspection (cont'd)

Examination or test	MIL-STD-750		LTPD	Symbol	Limits		Unit
	Method	Details			Min	Max	
Subgroup 8			λ10				
High temperature life (non-operating)	1031	T_A = 150° C		---	---	---	---
End points: (Same as for subgroup 7)							
Subgroup 9			λ10				
Operational life	(see 4.5)	Test circuit figure 6		---	---	---	---
		T_A = 100° C					
End points: (Same as for subgroup 7)							

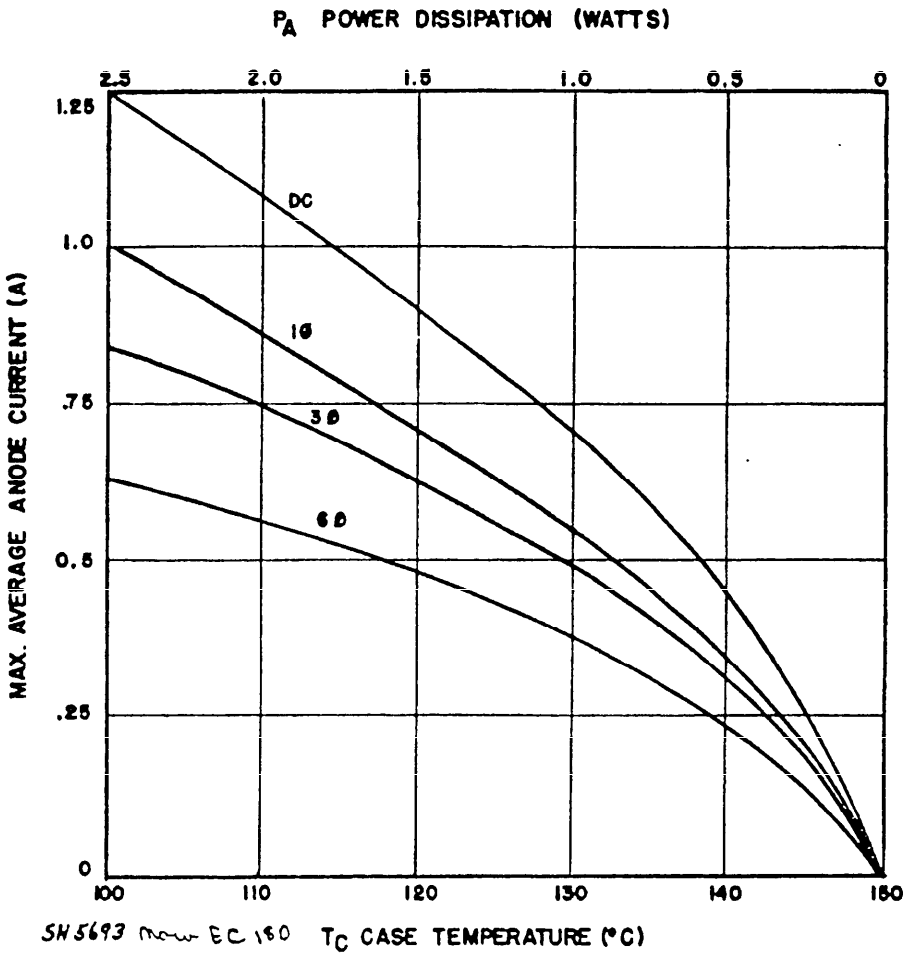
4.4 Holding current. The holding current is measured by decreasing the anode current to the point where the device under test switches from the on-state to the off-state. The gate current shall be -150 μA and the anode supply voltage shall be 5 volts. The test circuit shown on figure 5, or its equivalent, may be used. Resistor R_4 is increased, thereby decreasing device anode current, until the device switches to the off-state. The current measured by M_1 immediately prior to switching is the holding current. Switch S_1 is used to trigger the device to the on-state prior to making a holding current measurement.

4.5 Operational life test. The operational life test shall be performed in the test circuit shown on figure 6, or its equivalent. Transformer T_1 is adjusted such that the peak value of secondary voltage from T_3 equals the rated forward blocking voltage of the device under test. Transformer T_2 is adjusted such that meter M_2 reads 220 mA average rectified current. Transformer T_5 is selected to provide a maximum secondary peak voltage of 5 volts as read on meter M_3. The device under test shall be in an oven at an ambient temperature of 100° ±5° C.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-S-19500.

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NOTE:
1. Ratings apply for 50 to 400 Hz frequency.

Figure 1. Anode current rating as a function of operating case temperature.

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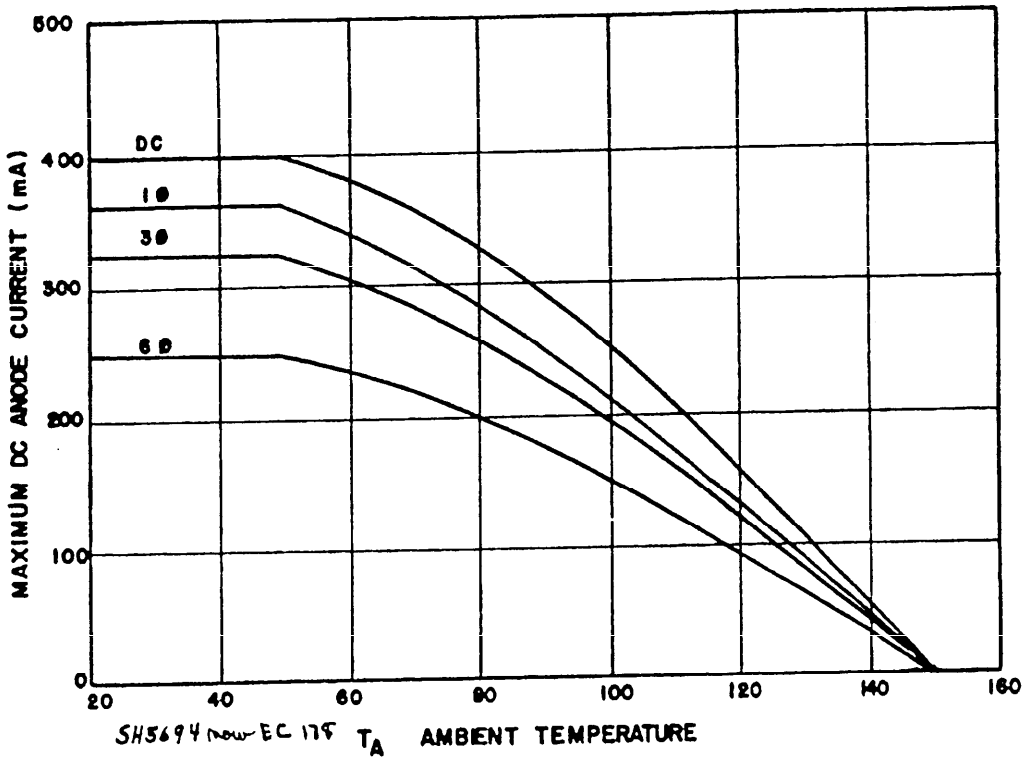


Figure 2. Anode current rating as a function of operating ambient temperature.

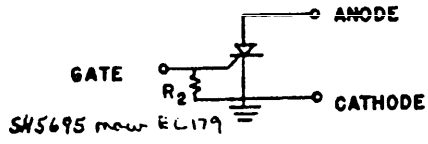
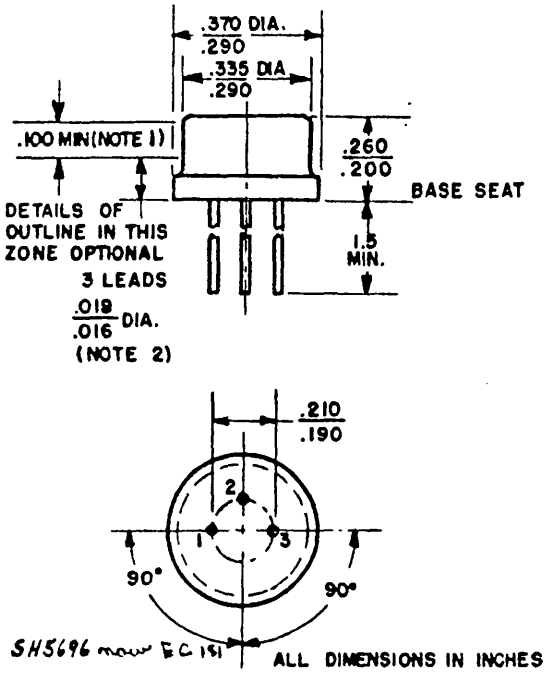


Figure 3. Condition for blocking voltage ratings.

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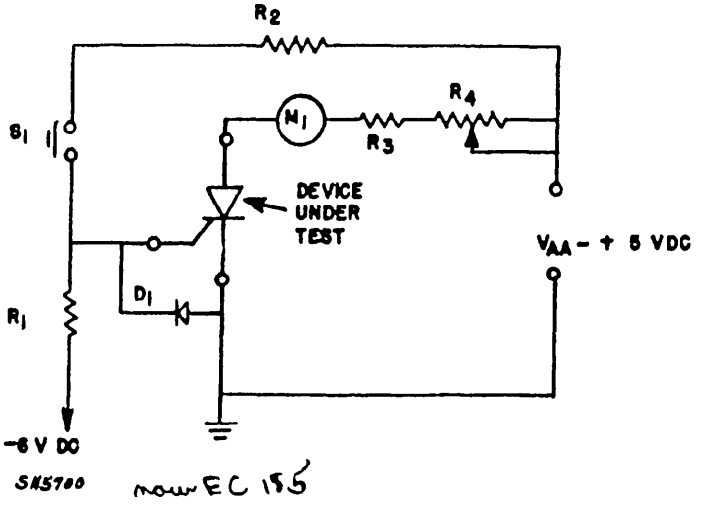
NOTES:

1. This zone is controlled for automatic handling. The variation in actual diameter within this zone shall not exceed 0.010.
2. The specified lead diameter applies in the zone between 0.050 and 0.250 from the base seat. Between 0.250 and 1.5 a maximum of 0.021 diameter is held. Outside of these zones the lead diameter is not controlled.
3. All terminals are electrically insulated from the case.
4. Leads may be soldered to within 1/16 inch of base. Heat sinking not required if temperature-time exposure is less than 230° C for 10 seconds.
5. This device is for socketed single-sided circuit-board, wire-in, and similar applications. Where used in double-sided or eyeletted circuit-board or similar applications where solder bridging may occur, a dielectric washer or other standoff device may be necessary.

Figure 4. Dimensions of semiconductor thyristors, types 2N1870A, 2N1871A, 2N1872A and 2N1874A.

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at voltage type represented in the lot may be used for Group B and C inspections as represented



M_1 - DC MILLIAMMETER

S_1 - TRIGGER ON SWITCH, MOMENTARY TYPE, NORMALLY OPEN

D_1 - IN457 DIODE

R_1 - 40 K OHMS, $\frac{1}{4}$ WATT

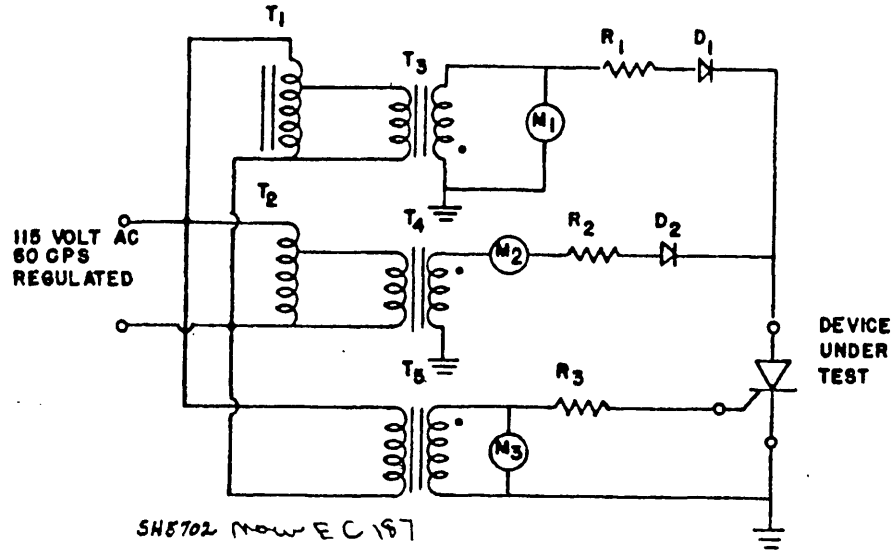
R_2 - 4700 OHMS, $\frac{1}{4}$ WATT

R_3 - 270 OHMS, $\frac{1}{4}$ WATT

R_4 - 25,000 OHM, POTENTIOMETER

Figure 5. Test circuit for holding current.

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T_1 – 115 VAC ADJUSTABLE TRANSFORMER

T_2 – 115 VAC ADJUSTABLE TRANSFORMER

T_3 – STEP UP TRANFORMER 115V TO 140 VAC

T_4 – STEP DOWN TRANSFORMER 115V TO 12 VAC

T_5 – STEP DOWN TRANSFORMER 115V TO 3.0 VAC

D_1 – IN540

D_2 – IN540

R_1 – 20,000 OHM, $\frac{1}{2}$ WATT

R_2 – 20 OHM, 20 WATT

R_3 – 1000 OHM, $\frac{1}{4}$ WATT

M_1 – PEAK READING VOLTMETER (OR RMS EQUIVALENT) 0-250 VOLTS

M_2 – DC MILLIAMMETER 0-300 mA

M_3 – PEAK READING VOLTMETER (OR RMS EQUIVALENT) 0-10 VOLTS

Figure 6. Circuit for operational life test.

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NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

Review activity:
Navy - SH

User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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